A First Course in Abstract Algebra

Section 3: Isomorphic Binary Structures

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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3 Isomorphic Binary Structures

Exercise 3.1: The three isomorphism checks

Check that the map is (1) one-to-one, (2) onto the specified target set, and (3) operation-preserving: $\phi(a * b) = \phi(a) *' \phi(b)$ for every $a, b \in S$. The first two make a bijection; the third ensures that it respects the algebraic structure. Also confirm initially that the formula actually takes values in the stated codomain.

Exercise 3.2: Negation on the integers

This is an isomorphism. If $-m = -n$ then $m = n$, proving injectivity. Each integer y is the image of $-y$, proving surjectivity. Finally $\phi(m+n) = -(m+n) = -m-n = \phi(m)+\phi(n)$, proving preservation of addition.

Exercise 3.3: Doubling integers

This map preserves addition since $2(m + n) = 2m + 2n$ and is injective because $2m = 2n$ implies $m = n$. However its image consists only of even integers; 1 has no preimage in $\mathbb{Z}$. Thus it is not onto and is not an isomorphism onto the specified target.

Exercise 3.4: Translating integers

The map $n \mapsto n + 1$ is bijective, with inverse $y \mapsto y - 1$, but does not preserve addition. Indeed $\phi(0 + 0) = 1$, while $\phi(0) + \phi(0) = 2$. A bijection alone is not an isomorphism.

Exercise 3.5: Halving rational numbers

If $x/2 = y/2$, then $x = y$, so the map is injective. A rational target q has rational preimage $2q$, so it is onto. Also $(x + y)/2 = x/2 + y/2$. All three requirements hold, making this an additive isomorphism.

Exercise 3.6: Squaring rational numbers

Squaring preserves multiplication: $(xy)^2 = x^2y^2$. It is not injective, since 1 and -1 have the same image, and is not onto $\mathbb{Q}$, since no rational square equals -1. Hence it is not an isomorphism.

Exercise 3.7: Cubing real numbers

The cubic function is strictly increasing, so it is injective, and every real y has real preimage $\sqrt[3]{y}$, so it is onto. Since $(xy)^3 = x^3y^3$, it preserves multiplication. Thus it is an isomorphism of the indicated multiplicative binary structures, even though the structures include zero and are not groups.

Exercise 3.8: Determinants of two by two matrices

Determinants preserve multiplication: $\det(AB) = \det(A)\det(B)$. The map is onto since $\det \operatorname{diag}(r, 1) = r$. It is not injective: the distinct matrices I_2 and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ both have determinant 1. Thus it is not an isomorphism.

Exercise 3.9: Determinants of one by one matrices

A one by one matrix $[r]$ has determinant r. The map $[r] \mapsto r$ is bijective, with inverse $r \mapsto [r]$. Also $[r][s] = [rs]$, whose determinant is rs. Consequently this determinant map is an isomorphism.

Exercise 3.10: A decreasing exponential

The codomain here is $\mathbb{R}^+$, the positive reals. The function $\phi(r) = (1/2)^r$ is strictly decreasing and has inverse $y \mapsto \log_{1/2} y$ for $y > 0$, so it is bijective. The exponent law gives $\phi(r+s) = (1/2)^{r+s} = \phi(r)\phi(s)$. Hence it is an isomorphism from addition to multiplication.

Exercise 3.11: Differentiating smooth functions

Differentiation preserves addition: $(f + g)' = f' + g'$. It is onto the smooth functions: if h is smooth, $f(x) = \int_0^x h(t)\,dt$ is smooth and has derivative h. But it is not injective, since the constant functions 0 and 1 both map to zero. Thus it is not an isomorphism.

Exercise 3.12: Evaluating a derivative at zero

The map is additive because $(f + g)'(0) = f'(0) + g'(0)$. It is onto $\mathbb{R}$, since $f(x) = cx$ has $f'(0) = c$ for any real c. It is not injective: distinct constant functions have the same image 0. Therefore it is not an isomorphism.

Exercise 3.13: Integration from zero

Integration is additive, and if $\int_0^x f(t)\,dt = \int_0^x g(t)\,dt$ for all x, differentiation gives $f = g$, proving injectivity. However each image function is zero at $x = 0$, so the constant function 1 is not in the image. Thus it is not onto F and is not an isomorphism.

Exercise 3.14: Differentiating the integral

The fundamental theorem of calculus gives $\phi(f)(x) = \frac{d}{dx}\int_0^x f(t)\,dt = f(x)$, since every smooth function is continuous. Thus the map is the identity on F, so is bijective and preserves addition. It is an isomorphism.

Exercise 3.15: Multiplication by the independent variable

The map is not onto: every image $xf(x)$ vanishes at zero, whereas the constant function 1 does not. It also fails to preserve multiplication: taking $f = g = 1$ gives $\phi(fg)(x) = x$ but $(\phi(f)\phi(g))(x) = x^2$, different at $x = 2$. Thus it is not an isomorphism for either reason.

Exercise 3.16: Transporting addition by a translation

The inverse of $\phi(n) = n + 1$ is $\phi^{-1}(u) = u - 1$.

(a) To send ordinary addition to $*$, define $u * v = \phi((u - 1) + (v - 1)) = u + v - 1$. Then $\phi(m) * \phi(n) = (m + 1) + (n + 1) - 1 = m + n + 1 = \phi(m + n)$. Its identity is $\phi(0) = 1$.

(b) To send $*$ to ordinary addition, require $\phi(m*n) = \phi(m)+\phi(n)$. This gives $m*n+1 = m+n+2$, or $m * n = m + n + 1$. Its identity is $-1 = \phi^{-1}(0)$.

Exercise 3.17: Transporting multiplication by a translation

Use $\phi(n) = n + 1$ and its inverse $u - 1$.

(a) Define $u * v = \phi((u - 1)(v - 1)) = (u - 1)(v - 1) + 1 = uv - u - v + 2$. Then $\phi(m) * \phi(n) = mn + 1 = \phi(mn)$. The identity is $\phi(1) = 2$.

(b) Require $\phi(m * n) = \phi(m)\phi(n)$, giving $m * n = (m + 1)(n + 1) - 1 = mn + m + n$. Its identity is $0 = \phi^{-1}(1)$.

These are binary operations on integers; the construction does not assert that every integer has a multiplicative inverse.

Exercise 3.18: Transporting rational addition

Here $\phi(x) = 3x - 1$ and $\phi^{-1}(u) = (u + 1)/3$.

(a) Define $u*v = \phi((u+1)/3+(v+1)/3) = u+v+1$. This satisfies $\phi(x)*\phi(y) = 3x+3y-1 = \phi(x+y)$. Its identity is $-1 = \phi(0)$.

(b) Define $x * y = \phi^{-1}(\phi(x) + \phi(y)) = (3x + 3y - 2 + 1)/3 = x + y - 1/3$. Its identity is $1/3 = \phi^{-1}(0)$.

All displayed formulas preserve rationality, so their domains and codomains are correct.

Exercise 3.19: Transporting rational multiplication

Again $\phi^{-1}(u) = (u + 1)/3$.

(a) Define $u * v = \phi(\phi^{-1}(u)\phi^{-1}(v)) = (u+1)(v+1)/3-1$. Then $\phi(x) * \phi(y) = 3xy - 1 = \phi(xy)$. Its identity is $\phi(1) = 2$.

(b) Define $x * y = \phi^{-1}(\phi(x)\phi(y)) = ((3x - 1)(3y - 1) + 1)/3 = 3xy - x - y + 2/3$. Its identity is $\phi^{-1}(1) = 2/3$.

In each direction the equation used to define the operation is exactly the required preservation condition, and the original map is already bijective.

Exercise 3.20: Commuting with the operations

There are two routes from an input pair (a, b) to an element of S'. One first computes $a * b$ in S and then applies ϕ. The other first applies ϕ to both inputs and then computes $\phi(a) *' \phi(b)$ in S'. Saying that ϕ commutes with these operations means precisely that the two routes give the same result. It does not mean ϕ itself is a binary operation.

Exercise 3.21: Completing the isomorphism definition

Let $*$ and $*'$ be binary operations on S and S'. An isomorphism $\phi : (S, *) \to (S', *')$ is a bijection such that $\phi(a * b) = \phi(a) *' \phi(b)$ for every $a, b \in S$. The original version omitted bijectivity and the universal quantifier.

Exercise 3.22: The identity definition

The stated definition is correct: an element $e \in S$ is an identity for $*$ when $s * e = s = e * s$ for every $s \in S$. The middle equality to s says multiplication by e leaves every element unchanged, and the two sides specify both a left and a right identity. Merely requiring $s * e = e * s$, without equality to s, would only require e to commute with every element and would not suffice.

Exercise 3.23: Synopsis of identity uniqueness

Compute the product of two proposed identities: using the first as a left identity and the second as a right identity makes that same product equal to each of them, forcing them to coincide.

Exercise 3.24: Left and right identities

(a) A left identity $e_L \in S$ satisfies $e_L * s = s$ for every $s \in S$. It need not be unique: on $S = \{a, b\}$ define $x * y = y$. Then both elements are left identities. In the two-sided uniqueness proof, the step $e_L * e'_L = e_L$ is unavailable: it would require e'_L to be a right identity.

(b) A right identity $e_R \in S$ satisfies $s * e_R = s$ for every $s \in S$. On $S = \{a, b\}$ define $x * y = x$. Then both elements are right identities. The step $e_R * e'_R = e'_R$ fails because it requires e_R to be a left identity.

Exercise 3.25: A left identity and a right identity together

They cannot differ. If l is a left identity and r a right identity, then $l * r = r$ by the left identity property and $l * r = l$ by the right identity property. Thus $l = r$, and this element is consequently a two-sided identity. No associativity assumption is needed.

Exercise 3.26: The inverse of an isomorphism

The inverse ϕ^{-1} is a bijection because ϕ is. Given $u, v \in S'$, write $a = \phi^{-1}(u)$ and $b = \phi^{-1}(v)$. Preservation by ϕ gives $\phi(a * b) = \phi(a) *' \phi(b) = u *' v$. Applying ϕ^{-1} yields

$$\phi^{-1}(u *' v) = a * b = \phi^{-1}(u) * \phi^{-1}(v).$$

Thus the inverse also preserves the operations, completing all requirements.

Exercise 3.27: Composing isomorphisms

A composite of two bijections is a bijection (its inverse is $\phi^{-1} \circ \psi^{-1}$). Also

$$(\psi \circ \phi)(a * b) = \psi(\phi(a) *' \phi(b)) = \psi(\phi(a)) *'' \psi(\phi(b)).$$

The first preservation step uses ϕ and the second uses ψ. This is precisely preservation by $\psi \circ \phi$, so the composite is an isomorphism.

Exercise 3.28: Isomorphism is an equivalence relation

Reflexivity: the identity map on any binary structure is bijective and operation-preserving. Symmetry: if one structure is isomorphic to another, the inverse isomorphism from Exercise 3.26 gives the reverse direction. Transitivity: compose the two isomorphisms using Exercise 3.27. Thus all three equivalence-relation conditions hold on any specified set of binary structures.

Exercise 3.29: Commutativity is structural

Suppose $\phi : S \to S'$ is an isomorphism and $*$ is commutative. For arbitrary $u, v \in S'$, surjectivity gives $u = \phi(a)$ and $v = \phi(b)$. Then

$$u *' v = \phi(a * b) = \phi(b * a) = v *' u.$$

Hence $*'$ is commutative. Applying the same argument to ϕ^{-1} proves the converse. Therefore commutativity depends on structure, not on the names of elements.

Exercise 3.30: Associativity is structural

For arbitrary $u = \phi(a), v = \phi(b), w = \phi(c)$ in S', preservation gives

$$(u *' v) *' w = \phi((a * b) * c) = \phi(a * (b * c)) = u *' (v *' w).$$

Surjectivity ensures every target triple has these preimages; associativity in S gives the middle equality. Repeat with the inverse to obtain the reverse implication. Thus associativity is invariant under isomorphism.

Exercise 3.31: Existence of square roots is structural

Assume each $c \in S$ has some x with $x * x = c$. Given $d \in S'$, choose $c = \phi^{-1}(d)$ and a square root x of c. Then $y = \phi(x)$ satisfies $y *' y = \phi(x * x) = \phi(c) = d$. The same argument using ϕ^{-1} proves the converse. We require existence, not uniqueness, of these roots.

Exercise 3.32: Existence of an idempotent is structural

If $b * b = b$ in S, then $\phi(b) *' \phi(b) = \phi(b * b) = \phi(b)$, so $\phi(b)$ is idempotent in S'. Conversely any target idempotent pulls back under ϕ^{-1} to an idempotent in S. Thus the existence of such an element is a structural property.

Exercise 3.33: Complex numbers as real matrices

Define $\phi(a+bi) = M(a,b) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. Equality of matrices recovers both a and b, proving injectivity, and every matrix in H has this form, proving surjectivity.

(a) $M(a,b) + M(c,d) = M(a+c, b+d) = \phi((a+bi)+(c+di))$, so the map preserves addition.

(b) $M(a,b)M(c,d) = M(ac-bd, ad+bc) = \phi((a+bi)(c+di))$, so it preserves multiplication.

The same bijection therefore gives both requested isomorphisms.

Exercise 3.34: Classifying two-element binary structures

Encode a table on $\{a,b\}$ by the four entries $(a*a, a*b, b*a, b*b)$. The only relabelings are the identity and the swap of a, b. Here are ten representative tables, specified by their four entries in that order:

	$a*a$	$a*b$	$b*a$	$b*b$
1	a	a	a	a
2	a	a	a	b
3	a	a	b	a
4	a	a	b	b
5	a	b	a	a
6	a	b	a	b
7	a	b	b	a
8	b	a	a	a
9	b	a	b	a
10	b	b	a	a

To see completeness without guessing, encode a, b as $0, 1$. A swap sends (t_1, t_2, t_3, t_4) to $(1-t_4, 1-t_3, 1-t_2, 1-t_1)$. Exactly four tables are fixed: choose t_1, t_2 freely and then $t_4 = 1-t_1, t_3 = 1-t_2$. The other twelve tables form six pairs. Hence there are $4 + 12/2 = 10$ classes; the ten displayed representatives lie in distinct swap classes and exhaust them.